

OPLOGS-01: OpenPipeline Fundamentals

Series: OPLOGS — OpenPipeline Logs | **Notebook:** 1 of 8 | **Created:** December 2025 | **Last Updated:** 09/24/2026

Understanding the Unified Data Ingestion Framework

This notebook introduces OpenPipeline, Dynatrace's unified data processing framework for logs, traces, metrics, and events.

 **New Addition (March 2026)**

Companion Series

- **OPMIG** — Migrating from classic logs to OpenPipeline? Start with **OPMIG-01: Why Migrate**
- **OPIPE** — Ready to process spans, metrics, and events? Continue to **OPIPE-01: OpenPipeline as a Multi-Scope Platform**
- **SPANS** — Need to query and analyze distributed traces? See **SPANS-01: Fundamentals**

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Prerequisites

-  Access to a Dynatrace environment with log data
-  DQL query permissions (viewer role minimum)
-  Basic understanding of log management concepts

1. What is OpenPipeline?

OpenPipeline is Dynatrace's unified data ingestion and processing framework that replaces classic log ingestion. It provides:

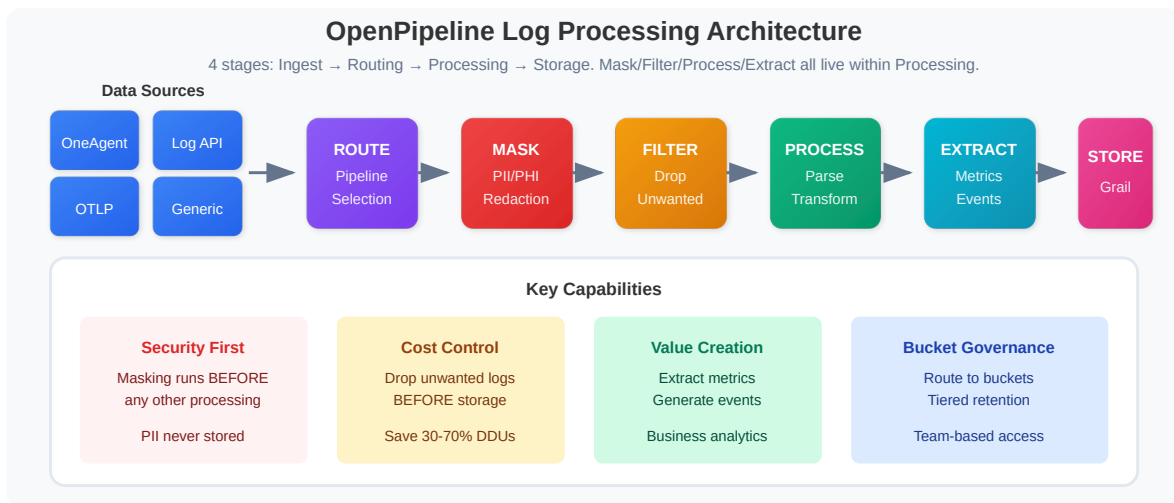
- **Unified Processing:** Single framework for logs, metrics, traces, and business events
- **Real-time Transformation:** Parse, enrich, mask, and route data at ingestion
- **Grail Storage:** Direct integration with Dynatrace's data lakehouse
- **Flexible Routing:** Send data to different buckets with custom retention
- **Cost Control:** Drop unnecessary data before storage

OpenPipeline vs Classic Log Ingestion

Feature	Classic Logs	OpenPipeline v2.0
Data Processing	Post-ingestion	At ingestion time
Storage	Log Storage v1	Grail Data Lakehouse
Query Language	Limited	Full DQL Support
Retention	Global	Per-bucket configurable
Data Masking	Limited	DQL processor with DPL (<code>replacePattern</code>), <code>replaceString</code> , <code>ipMask</code>

Feature	Classic Logs	OpenPipeline v2.0
Parsing	Basic	DPL (Dynatrace Pattern Language)
Custom Routing	No	Yes, by content/source

2. OpenPipeline Architecture



Stage model (verified 09/24/2026): data flows Ingest → Routing → pipeline → Storage. Inside a pipeline, per *Processing in OpenPipeline*, "The sequence of stages is fixed for all pipelines and cannot be modified." Masking, dropping, parsing and transformation are processors in the first stage, **Processing**; **Bucket assignment** comes seventh, and **Metric extraction**, **Davis** and **Data extraction** run after it. The diagram's Mask / Filter / Process boxes are all inside that first stage.

Sources: [Processing in OpenPipeline \(DT docs\)](#)

3. Exploring Your OpenPipeline Data

Let's start by discovering what data sources and pipelines are active in your environment.

```
// Discover data sources feeding OpenPipeline
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {dt.openpipeline.source}
| sort log_count desc
```

```
// See which pipelines are processing your logs
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {dt.openpipeline.pipelines}
| sort log_count desc
```

```
// Check storage bucket distribution
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {dt.system.bucket}
| sort log_count desc
```

4. Key OpenPipeline Fields

OpenPipeline adds metadata fields to every log record:

Pipeline Metadata

Field	Description	Example
<code>dt.openpipeline.source</code>	How the log was ingested	<code>oneagent</code> , <code>/api/v2/logs/ingest</code> , <code>/api/v2/otlp/v1/logs</code>
<code>dt.openpipeline.pipelines</code>	Pipeline(s) that processed the log	<code>["logs:pipeline_Default_Pipeline_2798"]</code>
<code>dt.system.bucket</code>	Storage bucket name	<code>default_logs</code> , <code>custom_logs</code>

Core Log Fields

Field	Description
<code>timestamp</code>	When the log was generated
<code>content</code>	The log message body
<code>loglevel</code>	Log severity (ERROR, WARN, INFO, DEBUG, NONE)

Field	Description
<code>status</code>	Severity group derived from <code>loglevel</code> : ERROR (SEVERE, ERROR, CRITICAL, ALERT, FATAL, EMERGENCY), WARN, INFO (INFO, TRACE, DEBUG, NOTICE), NONE — use it for error-class counts
<code>log.source</code>	Source identifier (e.g., "Container Output")
<code>log.iostream</code>	Stream type (stdout, stderr)

```
// View a sample log record with all key fields
fetch logs, from: now() - 1h
| fields timestamp, content, loglevel, status, log.source, log.iostream,
      dt.openpipeline.source, dt.openpipeline.pipelines, dt.system.bucket
| limit 5
```

```
// Analyze log levels in your environment
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {loglevel}
| sort count desc
```

5. Data Sources Explained

OneAgent (`oneagent`)

Logs collected automatically by Dynatrace OneAgent from:

- Container stdout/stderr
- Process log files
- System logs

Log Ingest API (`/api/v2/logs/ingest`)

Logs sent directly via the Dynatrace API:

- Custom application logs
- Third-party integrations
- Cloud provider logs (AWS, Azure, GCP)

Delivery reliability

The Log Ingest API acknowledges accepted data with **HTTP 204 No Content** (empty body). When a request fails on a *retryable* response code (each endpoint documents which codes

are retryable), the **client is responsible for retrying with exponential backoff** — Dynatrace does not retry on the sender's behalf. The ActiveGate that serves the endpoint buffers ingested data to an on-disk queue (default **300 MB**, configured by `disk_queue_max_size_mb`; location by `disk_queue_path`) and forwards it to Dynatrace in batches; when that queue fills, the API returns **503 Usable space limit reached** as a back-pressure signal — increase `disk_queue_max_size_mb` if this recurs under normal load. Cloud-forwarder integrations (e.g. Amazon Data Firehose) implement their own buffering and smart-retry against the OpenPipeline endpoint.

OTLP (`/api/v2/otlp/v1/logs`)

OpenTelemetry Protocol logs:

- OpenTelemetry Collector
- Fluent Bit with OTLP output
- Custom OTLP exporters

```
// Compare volume by data source
fetch logs, from: now() - 24h
| summarize {
    log_count = count(),
    unique_hosts = countDistinct(dt.entity.host)
}, by: {dt.openpipeline.source}
| sort log_count desc
```

```
// Logs per hour by source (trend analysis)
fetch logs, from: now() - 24h
| makeTimeseries {log_count = count()}, by: {dt.openpipeline.source}, interval:
1h
```

6. Pipeline Stages Overview

A record first passes **Routing**, which picks the pipeline (a dynamic DQL matcher, or a static route for custom ingest sources). Inside the pipeline it then runs through a **fixed** sequence of ten stages — the order cannot be changed:

#	Stage	What it does in a log pipeline	Processors	Executed
1	Processing	Parse values into fields, transform the schema, filter records, mask sensitive data	DQL, Add fields, Remove fields, Rename fields, Drop record, GeoIP lookup (Early Access), Inline lookup	All matches
2	Smartscape node	Extract Smartscape nodes from matching records	Smartscape node	All matches
3	Smartscape edge	Extract Smartscape edges from matching records	Smartscape edge	All matches
4	Permission	Apply a security context to matching records	Set security context	First match only
5	Product allocation	Assign product or application usage	DPS Cost Allocation - Product	First match only
6	Cost allocation	Assign cost-center usage	DPS Cost Allocation - Cost Center	First match only
7	Bucket assignment	Choose the Grail bucket — or not to store the record	Bucket assignment, No storage assignment	First match only
8	Metric extraction	Extract metrics from matching records	Counter metric, Histogram metric, Value metric	All matches
9	Davis	Extract a Davis event and re-ingest it into another pipeline	Davis event	All matches
10	Data extraction	Extract a business event or SDLC event and re-ingest it	Business event, Software development lifecycle event	All matches

Masking, dropping and parsing are not separate stages: they are processors in stage 1, and within a stage they run in the order you place them. Storage in the chosen Grail bucket happens after the pipeline, so a record's retention is set by stage 7 — which is also why the extraction stages (8–10) still see records assigned to **No storage assignment**.

Sources: [Processing in OpenPipeline \(DT docs\)](#) — "each processor output becomes the input for the next one", [OpenPipeline processing examples \(DT docs\)](#) — "The record continues through all pipeline stages,

including metric extraction, and is not stored only at the end."

7. Environment Summary

Let's get a complete picture of your OpenPipeline environment:

```
// Complete environment summary
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    unique_sources = countDistinct(dt.openpipeline.source),
    unique_buckets = countDistinct(dt.system.bucket),
    unique_hosts = countDistinct(dt.entity.host),
    error_count = countIf(loglevel == "ERROR" OR status == "ERROR"),
    warn_count = countIf(loglevel == "WARN" OR status == "WARN")
}
```

```
// Top log sources by entity
fetch logs, from: now() - 1h
| filter isNotNull(dt.entity.host)
| summarize {log_count = count()}, by: {host.name, log.source}
| sort log_count desc
| limit 15
```



Summary

In this notebook, you learned:

- ✓ **What OpenPipeline is** - Dynatrace's unified data processing framework
- ✓ **Architecture** - Data flow from sources through processing to Grail
- ✓ **Key fields** - `dt.openpipeline.source`, `dt.openpipeline.pipelines`, `dt.system.bucket`
- ✓ **Data sources** - OneAgent, Log Ingest API, OTLP
- ✓ **Pipeline stages** - a fixed sequence: Processing (mask, drop, parse, transform) → ... → Bucket assignment → Metric / Davis / Data extraction



Next Steps

Continue to **OPLOGS-02: Migration Guide** to learn how to migrate from classic log ingestion to OpenPipeline v2.0.



References

- [OpenPipeline \(DT docs\)](#)
 - [Processing in OpenPipeline \(DT docs\)](#)
 - [Grail \(DT docs\)](#)
 - [Dynatrace Query Language \(DT docs\)](#)
 - [Automatic log enrichment \(DT docs\)](#) — "a status attribute is created with a value that is a sum of loglevel values based on the following grouping"
 - [Log ingestion API \(DT docs\)](#) — delivery reliability: HTTP 204 success, retryable codes + exponential backoff, ActiveGate disk-queue buffering, 503 Usable space limit reached
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